

THE EVALUATION OF INCIDENTALLY DISCOVERED ADRENAL MASSES

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ABSTRACT

Objective: The objective of this Disease State Clinical Review is to provide clinicians with a practical approach to the evaluation of incidentally discovered adrenal masses.

Methods: A case-based clinical approach to the evaluation of adrenal masses is presented. Recommendations were developed using available prospective and randomized studies, cohort studies, cross-sectional studies, anecdotal observations, and expert opinions.

Results: Incidentally discovered adrenal masses are common. The approach to the patient with an adrenal mass should involve assessment of malignant potential via imaging characteristics and adrenal hormone excess via clinical and biochemical features. The roles of biopsy, surgical or medical therapy, and longitudinal surveillance are also important to consider and are influenced by case-specific factors. Inappropriate or inadequate evaluations may put patients at increased risk for developing preventable adverse cardiometabolic outcomes or cancer.

Conclusion: Incidentally discovered adrenal masses require a multimodal assessment that involves interpretation of multiple imaging characteristics, dynamic and static hormonal measurements, and a nuanced approach to considering interventional diagnostics, treatments, and longitudinal surveillance. Herein, we review these evaluations and provide a practical approach for clinicians. (*Endocr Pract.* 2019;25:178-192)

Abbreviations:

ACTH = adrenocorticotropic hormone; CS = Cushing syndrome; CT = computed tomography; DHEAS = dehydroepiandrosterone sulfate; HU = Hounsfield units; MRI = magnetic resonance imaging; UFC = urinary free cortisol

INTRODUCTION

The prevalence of incidentally discovered adrenal masses (also referred to as adrenal incidentalomas or adrenal nodules) has grown substantially with the increased use of cross-sectional imaging such as computed tomography (CT) and magnetic resonance imaging (MRI). Adrenal masses may be discovered incidentally during emergency room visits or during the course of ambulatory or hospitalization-related imaging of the abdomen or chest. Increasingly, adrenal masses may also be discovered during the course of surveillance imaging associated with cancer care, though these findings are not truly incidental (1). Though adrenal masses, or adrenal hyperplasia or thickening, of any size may have clinical relevance, this review will focus on the evaluation of incidentally discovered adrenal masses greater than or equal to 1 cm in size.

The true prevalence of adrenal masses is not known; rather, estimates are extrapolated from limited observations. The prevalence of incidental adrenal masses based

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on 25 postmortem series is about 6% (range, 1.1 to 32%) (2-5). The prevalence of incidentally discovered adrenal masses using high-resolution CT scans is slightly lower, at approximately 4% (2-4,6-9) and generally observed to be higher with older age (2-4,6,7). Given that the annual frequency of CT and MRI imaging studies exceeds 100 million per year in the United States, a statistic that has been steadily increasing in the last decade, the absolute number of annually discovered adrenal masses is expected to be large (10). As modern medicine continues to increase its reliance on abdominal imaging techniques, there will likely be a parallel increase in the incidental discovery of adrenal masses. Unfortunately, our understanding of risk factors for developing adrenal masses, and our ability to predict either their presence and/or outcomes after diagnosis, is limited. Clinicians will be faced with the challenge of evaluating these adrenal masses to ensure that they do not represent health hazards or harbingers of future risk. This American Association of Clinical Endocrinologists Disease State Clinical Review aims to provide practicing clinicians with a simple and case-based overview on approaching incidentally discovered adrenal masses and is presented in a unique patient-oriented format that builds upon prior guidelines and consensus statements (1,11-14).

The two fundamental questions each clinician is faced with when an adrenal mass is identified are:

1. Does this adrenal mass represent a malignancy?
2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

Other specific questions may also arise in the context of each patient's specific situation, for example:

3. Would a biopsy of the adrenal mass aid with the diagnosis, management, or prognosis?
4. Is there an indication for surgical or medical treatment?
5. Is there any indication for longitudinal surveillance with imaging and/or biochemical testing? If so, how frequently and for what duration?

Herein, we will review the approach to answering these key questions by providing five patient scenarios, each selected to highlight a unique and relatively characteristic presentation that clinicians may encounter. It should be noted that this review is intentionally intended to serve as a *brief and practical companion* for the busy clinician rather than a formal "guideline;" recent comprehensive guidelines on this subject are available (13) and can provide more detailed evidence and expert opinions. Further, clinical practice recommendations should rely on the highest grades of evidence; however, since there are few randomized controlled trials and longitudinal cohort studies to dictate the most efficacious and cost-effective evidence-based approach to managing incidentally discovered adrenal masses, *this review relies on lower grades of*

evidence (some cohort studies, but mostly retrospective and cross-sectional studies, anecdotal observations, and expert opinion).

It is with these limitations that these recommendations are presented as a guiding framework for clinical care that will undoubtedly evolve in the future, and with the knowledge that each clinician must ultimately rely on their personal judgment, and the informed opinion of their individual patients, when pursuing collective decision-making.

CASE 1: *A 50-year-old woman presented to the emergency department with worsening left lower-quadrant pain. An abdominal CT scan with intravenous contrast was performed to evaluate for diverticulitis. No gastrointestinal abnormalities were seen; however, a right adrenal mass was noted. She was seen by her primary care doctor 1 month later. She was normotensive, normokalemic, and had no signs or symptoms of Cushing syndrome (CS). A repeat CT scan without intravenous contrast was ordered to specifically evaluate the right adrenal mass, which was noted to be 1.0 cm in size and described as being round and homogenous with an unenhanced attenuation of 8 Hounsfield units (HU) (Fig. 1). The left adrenal gland was normal. A 1-mg dexamethasone suppression test was performed, yielding a morning cortisol of 1.3 µg/dL.*

1. Does this adrenal mass represent a malignancy?

No, the radiographic characteristics of this patient's incidentally discovered right adrenal mass are suggestive of a benign adrenocortical adenoma.

Adrenal masses may be benign or malignant and hormonally functional or nonfunctional. Most incidentally discovered adrenal masses will represent a benign entity, of which the most common etiology is by far an adrenocortical adenoma (Table 1). Although primary adrenal malignancy (adrenocortical carcinoma) is very rare, the prevalence of metastases to the adrenal from extra-adrenal malignancies is not uncommon and may be increasing as cancer survival improves. The determination of malignant potential is usually a function of the radiographic appearance and size of the adrenal mass. Adrenocortical adenomas are usually lipid-rich, a feature that classically manifests as a low attenuation on unenhanced CT (<10 HU) and/or the presence of chemical shift on MRI (Table 2) (15-21).

In cases when the unenhanced CT attenuation is not characteristically lipid rich (>10 HU), a benign lipid-poor adrenocortical adenoma is statistically still the most likely diagnosis. In these instances, in- and out-of-phase imaging on MRI can be performed to obtain additional information, but may provide similar information, since chemical shift also assesses lipid content. Alternatively, CT imaging with a contrast washout protocol may provide additional diagnostic value. The most widely used washout protocol entails obtaining an unenhanced CT image at base-

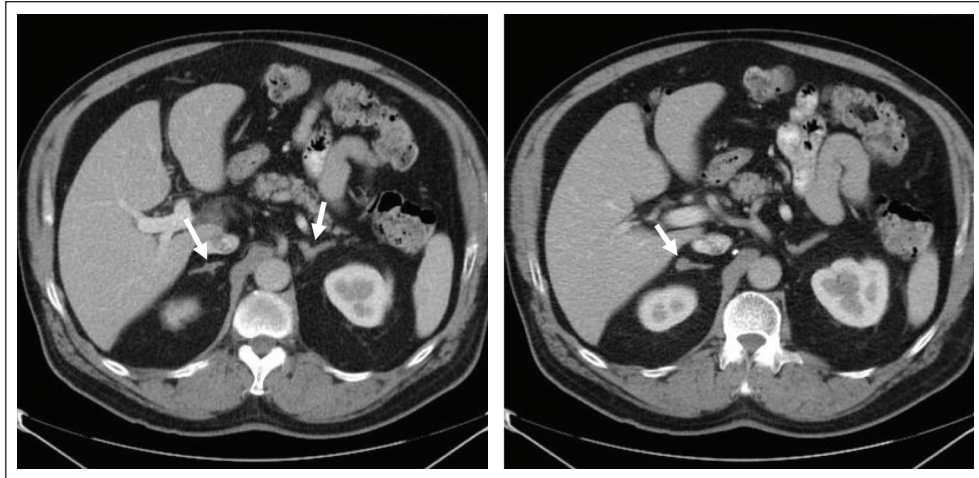


Fig. 1. Noncontrast axial computed tomography images of patient in Case 1 showing section with normal-appearing left and right adrenal glands (left image and arrows) and section highlighting 1.0-cm right adrenal mass (right image and arrow).

Table 1 The Differential Diagnosis of an Incidentally Discovered Adrenal Mass		
	Nonfunctional	Functional
BENIGN	Adrenocortical adenoma Cyst Ganglioneuroma Hemangioma Hemorrhage Infections and granulomatous disease (tuberculosis, fungi, sarcoidosis) Lymphangioma Myelolipoma Pheochromocytoma Schwannoma	Adrenocortical adenoma <i>Aldosterone producing</i> <i>Cortisol producing</i> <i>Androgen producing</i> ^a Micro- or Macronodular disease <i>Aldosterone producing</i> <i>Cortisol producing</i> Pheochromocytoma Myelolipoma ^a Ganglioneuroma ^a
MALIGNANT	Adrenocortical carcinoma Metastatic cancer from a nonadrenal primary neuroblastoma	Adrenocortical carcinoma Pheochromocytoma
^a Rare instances of functional myelolipomas and ganglioneuromas and benign androgen-secreting adenomas have been described.		

line, followed by administration of intravenous contrast and repeat imaging at 1 minute and again at 15 minutes. Benign and lipid-rich adenomas are not contrast avid and will usually exhibit a >60% absolute washout of contrast after 15 minutes, with a >40% relative washout (Table 2) (22,23). When considering which imaging modality to use, multiple factors can be weighted. Most data and experience with characterizing adrenal masses has been with CT; however, CT imaging exposes patients to radiation. MRI is an alternative with no radiation exposure; however, the procedure takes longer than CT and is more costly. It is important to note that high-quality evidence to support the most effective imaging approach and modality is lacking, and therefore, these recommendations are based on expert opinion and reasonable clinical judgement (23-28). Other imaging characteristics that should raise concern for a

potential malignancy include: fluorodeoxyglucose (FDG) avidity on positron emission tomography, irregular (not round) shape, calcifications, and a rapid rate of growth if serial imaging is available (Table 2). Finally, large size (>4 cm) is sometimes used as a crude indicator of a potentially malignant or concerning lesion; however, it should be noted that malignant lesions may present as small in size, and benign lesions may present as large masses (29). Therefore, other radiographic characteristics (Table 2) are usually more specific indicators than size alone.

2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

This patient exhibits no evidence of clinically relevant hypercortisolism. There is no strong indication to

Table 2
Radiographic Characteristics Suggestive of Benign Versus Malignant Adrenal Masses

Characteristics	Likely benign	Potentially malignant
Irregular shape	No	Yes
Heterogeneous content	No	Yes
Necrosis or calcifications	No	Yes
Rate of growth	<1 cm/year	≥1 cm/year
Attenuation on unenhanced CT	<10 HU	≥10 HU
Contrast washout on CT protocol at 15 minutes	Absolute >60% Relative >40%	Absolute ≤60% Relative ≤40%
MRI chemical shift suggestive of lipid-rich content	Yes	No
FDG avidity on PET	No	Yes
Size	<4 cm	≥4-6 cm
Abbreviations: CT = computed tomography; FDG = fluorodeoxyglucose; HU = Hounsfield units; PET = positron emission tomography.		

screen for primary aldosteronism in the absence of hypertension or hypokalemia, and the imaging characteristics are not suggestive of pheochromocytoma.

Even when radiographic characteristics support an adrenal mass as being benign, a biochemical evaluation for adrenal hormone excess should be considered. Since most adrenal masses are detected incidentally, these patients usually do not exhibit symptoms or signs of an overt syndrome of adrenal hormone excess such as CS, primary aldosteronism, or pheochromocytoma. However, even mild or autonomous hypercortisolism (discussed in more detail in Case 3) can increase the risk for future cardiometabolic and skeletal risk (30-35). Accordingly, it is generally recommended that all patients with an adrenal mass be evaluated with an overnight 1-mg dexamethasone suppression test, where postdexamethasone morning cortisol values ≤1.8 µg/dL are considered to be “normal” or indicative of “nonfunctional” status (13). Patients with hypertension and/or hypokalemia should be screened for primary aldosteronism (36), and screening for pheochromocytoma should be considered in all patients with a lipid-poor adrenal mass, since it is important to detect pheochromocytoma prior to the development of overt adrenergic symptoms (Table 3). Patients with a homogenous, non-contrast avid, and lipid-rich adrenal mass (<10 HU) do not require biochemical evaluation for pheochromocytoma because the case detection in such scenarios is close to zero and complicated by a relatively high rate of false-positive biochemical testing (37-41).

3. Would a biopsy of the adrenal mass help with diagnosis or management?

There is no role for an adrenal biopsy in this case.

The role of adrenal mass biopsy is limited. A biopsy should be considered when an extra-adrenal metastasis

or an infection are the suspected etiology of the adrenal mass(es) and when the information from cytology is likely to influence management. Extra-adrenal metastases should be suspected when a patient has a history of nonadrenal cancer, when there are bilateral adrenal masses, and the radiographic appearance of the masses is dense and vascular and/or with an irregular shape (classically, neither round nor with smooth contour). In these situations, a biopsy can determine the type and stage of the primary cancer, which may influence subsequent management. A biopsy may also be considered when an infiltrative infection to the adrenal glands is suspected, such as by *Mycobacterium* or fungal organisms, and there are no other means to confirm the diagnosis. Patients with infectious adrenal infiltrations are often partially immunocompromised or had lived or traveled in potentially endemic regions. A biopsy in this situation may inform therapy towards specific antimicrobials rather than surgical procedures.

A biopsy is not recommended to distinguish a primary adrenocortical adenoma from a carcinoma; biopsy in these situations may be inaccurate and/or falsely reassuring due to sampling bias, and there is a theoretical risk of tumor seeding the needle track (42-44). When concern for a primary adrenal malignancy is raised based on radiographic imaging characteristics, a cancer-focused surgery should be strongly considered for diagnosis, prognosis/staging, and treatment. Infrequently, a suspected adrenal malignancy is deemed to be surgically unresectable, yet confirmation of the malignancy via histopathology is necessary to determine appropriate medical therapy (i.e., chemotherapy). In this circumstance, a biopsy of a likely malignant mass may be considered. Biopsy is contra-indicated in any mass which could represent a pheochromocytoma due to concern of precipitating a catecholamine crisis (45); rather, biochemical diagnosis, pre-operative adrenergic receptor blockade, and surgery should be pursued.

Table 3
Recommended Biochemical Tests to Evaluate for Adrenal
Hormone Excess in a Patient With an Incidentally Discovered Adrenal Mass

Condition	Patients to Test	Test	Abnormal Value
Autonomous cortisol secretion	All	1-mg Dexamethasone suppression test	Nonfunctional: ≤ 1.8 $\mu\text{g/dL}$ Possibly functional: 1.9-5 $\mu\text{g/dL}$ Autonomous hypercortisolism: >5.0 $\mu\text{g/dL}$
Primary aldosteronism	Hypertension and/or hypokalemia	Serum aldosterone to plasma renin activity ratio (ARR)	Suppressed renin ARR >20 -30
Pheochromocytoma	Lipid-poor, contrast-avid, heterogeneous adrenal masses	Plasma (or urinary) fractionated metanephrines	Greater than 2- to 4-fold higher than the upper limit of the reference range
Adrenal androgen excess	Hirsutism or virilization	DHEAS Total testosterone	Higher than the upper limit of the reference range

Abbreviation: DHEAS = dehydroepiandrosterone sulfate.

4. Is there an indication for surgical or medical treatment?

There is no indication for surgical or medical treatment in this case, since the adrenal mass is neither suspicious for malignancy nor hormonally active.

The general approach to the incidentally discovered adrenal mass is outlined in Figure 2. History and physical examination should first be conducted to assess suspicion for an obvious syndrome of adrenal hormone excess, followed by biochemical testing for evidence of adrenal hormone excess. If hormone excess is confirmed, localization of the source and surgical resection should be strongly considered (Fig. 2).

Most patients with an incidentally discovered adrenal mass are asymptomatic and do not exhibit an obvious syndrome of adrenal hormone excess and have normal values on biochemical screening. In these instances, the radiographic characteristics of the adrenal mass dictate subsequent management. Adrenal masses that display suspicious or concerning features for a potential malignancy can be further characterized using alternative imaging modalities if needed, and if unilateral, surgically removed. When there is uncertainty regarding malignant potential or ambiguous characterization that prevent firm decision making, repeated imaging can be performed in 3 to 6 months to reassess for progressive radiographic or biochemical features that may confirm suspicions of a potential malignancy (Fig. 2).

5. Is there any indication for longitudinal surveillance with imaging and/or biochemical testing? If so, how frequently and for what duration?

There is no strong evidence supporting longitudinal biochemical or radiographic surveillance for this patient.

There is a lack of large and robust prospective studies that evaluate the evolution of biochemical and radiographic characteristics of benign-appearing nonfunctional adrenal masses. The summary of multiple smaller and retrospective studies suggests that a nonfunctional and lipid-rich appearing adrenal mass that is <4 cm in size is extremely unlikely to represent a malignancy, and therefore, there is no strong evidence to repeat imaging studies (46,47). While there are rare reported cases where an adrenocortical carcinoma developed in a patient with a prior history of benign adrenocortical adenoma based on imaging criteria (48), it is not clear if the adrenocortical carcinoma in such cases developed from the adenoma or was an independent occurrence. Furthermore, it is not known whether imaging surveillance would have changed the clinical outcome in these rare scenarios. Accordingly, the authors do not feel that such rare cases should change imaging surveillance strategies for millions of benign-appearing adenomas until a better understanding of adrenocortical carcinoma pathogenesis and development is available.

Development of mild or autonomous hypercortisolism (previously referred to as “subclinical hypercortisolism” or “subclinical Cushing syndrome”) is the most commonly reported hormonal excess during long-term follow-up of adrenal incidentalomas (49); however, robust data to support serial biochemical evaluations are lacking. Rather, an individualized approach should be taken for each patient. Patients, and their primary care clinicians, should be educated about potential signs and symptoms of hypercortisolism and hyperaldosteronism that should trigger re-evaluation. The development of new or unexpectedly worsening hyperglycemia, diabetes, hypertension, hypokalemia, weight gain or lipodystrophy, and/or low bone mineral density or fractures in this patient may raise concern for hypercortisolism or hyperaldosteronism, and biochemical screening can be repeated (Fig. 2).

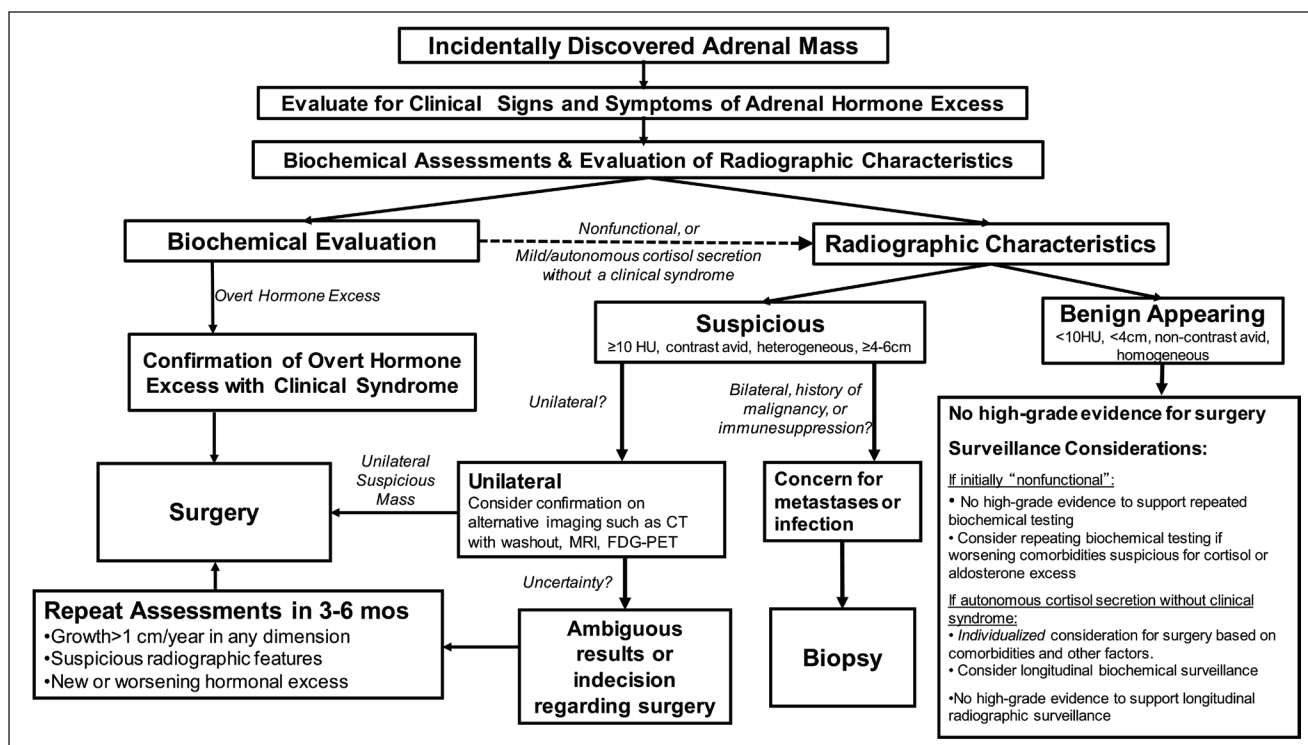


Fig. 2. Simplified approach to the incidentally discovered adrenal mass. CT = computed tomography; FDG-PET = fluorodeoxyglucose–positron emission tomography; HU = Hounsfield units; MRI = magnetic resonance imaging.

CASE 2: A 71-year-old woman presented to the emergency department with right flank pain. An abdominal CT scan was performed without contrast to evaluate for nephrolithiasis. Imaging revealed a small nonobstructing kidney stone and an incidentally observed $3.8 \times 2.9 \times 1.9$ cm diffuse left adrenal thickening/mass that was poorly circumscribed, had a 25 HU unenhanced attenuation with an absolute contrast washout of 30% after 15 minutes. In addition, the right adrenal gland was noted to be diffusely thickened without a discrete mass (Fig. 3). There were no other abdominal abnormalities, and subsequent imaging of the chest showed no abnormalities.

The patient had a history of breast cancer diagnosed 10 years earlier (positive for the estrogen and progesterone receptors and negative for Her2/neu) that was treated with lumpectomy, chemotherapy, and subsequently 7 years of aromatase inhibitor therapy. She had undergone surveillance imaging annually until 2 years prior to the current presentation. Prior imaging of her abdomen 2 years earlier revealed normal adrenal glands; thus, the current adrenal abnormalities were new.

The patient denied any symptoms of a pheochromocytoma, CS, or hyperandrogenism, and had no hypertension or hypokalemia. Plasma metanephrines were normal, and a 1-mg dexamethasone suppression test resulted in a morning cortisol of $0.9 \mu\text{g/dL}$.

1. Does this adrenal mass represent a malignancy?

These imaging characteristics are concerning for a malignancy.

The unenhanced attenuation of 25 HU suggests a lipid-poor entity. Further, low absolute washout percentage, the lack of smooth contours and irregular shape, bilateral adrenal involvement, and the background of a prior history of breast cancer not only raise concern for a malignant process but raise the specter of adrenal metastases. Metastases to the adrenal glands can be bilateral and may appear dense and irregular in shape on imaging. A primary adrenal malignancy or infection are less likely scenarios.

2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

There is no clinically apparent evidence of hypercortisolism, pheochromocytoma, androgen-secreting tumor, or primary aldosteronism. Furthermore, a 1-mg dexamethasone suppression test and plasma metanephrines were normal.

3. Would a biopsy of the adrenal mass help with diagnosis or management?

Yes, an adrenal biopsy may influence management decisions.

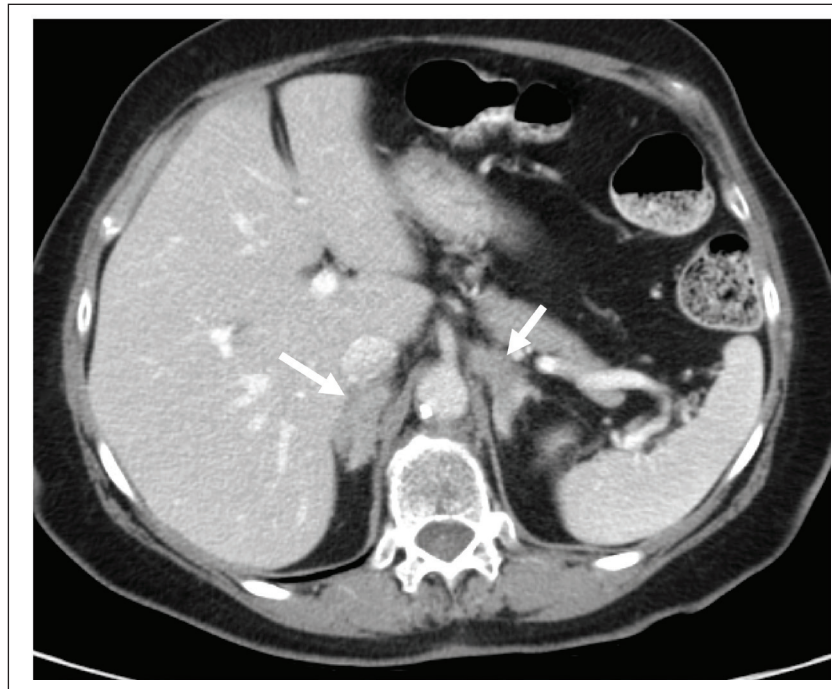


Fig. 3. The postcontrast transverse computed tomography (CT) image of Case 2 showing a $3.8 \times 2.9 \times 1.9$ cm left adrenal mass that is poorly circumscribed with an unenhanced CT attenuation of 25 Hounsfield units and 30% absolute washout at 15 minutes; there is diffuse thickening of right adrenal gland as well.

Since the primary consideration is metastatic breast cancer and there are no other metastatic lesions to sample, an adrenal biopsy could provide a tissue diagnosis which would also confirm the disease as advanced or stage 4. After excluding a pheochromocytoma, a biopsy may influence whether the appropriate treatment should involve systemic medical therapy versus surgical therapy (42).

In this case, a biopsy of the left adrenal mass revealed malignant cells consistent with breast cancer that was positive for the estrogen and progesterone receptors and negative for Her2/neu. This finding confirmed that the patient had a recurrence of her breast cancer presenting with isolated adrenal metastases nearly 10 years after her original disease.

4. Is there an indication for surgical treatment?

Surgery to remove an adrenal metastasis is reasonable to consider. However, in this case, the findings suggest that medical therapy for metastatic breast cancer might be a potential option.

This patient was resumed on an aromatase inhibitor, and 1 year later, imaging revealed complete regression of her adrenal metastases.

Although medical therapy was effective in this case, surgical metastasectomy should be considered on a case-by-case basis, especially when a patient's malignancy

is considered unlikely to respond to chemotherapies or targeted treatments.

5. Is there an indication for longitudinal surveillance with imaging and/or biochemical testing?

Yes, this patient should be closely monitored for primary adrenal insufficiency.

Given bilateral adrenal metastases, it would have been reasonable to evaluate adrenocorticotrophic hormone (ACTH) and cortisol levels at the time of diagnosis of metastatic breast cancer to establish a baseline with which to compare subsequent assessments. This patient was followed for signs and symptoms of primary adrenal insufficiency and routine morning cortisol and ACTH levels. Approximately 1 year after resuming the aromatase inhibitor, her morning cortisol levels declined and her ACTH levels rose, indicating the development of primary adrenal insufficiency. She was treated with replacement hydrocortisone and fludrocortisone.

CASE 3: *A 68-year-old man with a history of osteoporosis, hyperlipidemia, and prediabetes was incidentally found to have a right-sided lipid-rich adrenal mass (<10 HU on unenhanced CT imaging), size $4.6 \times 3.5 \times 3.5$ cm, on a CT scan performed to evaluate back pain (Fig. 4). The CT scan also noted vertebral compression*

fractures of his spine. Physical exam was unremarkable and there was no evidence of supraclavicular fat pads, striae, or proximal muscle weakness. Biochemical evaluation excluded pheochromocytoma and primary hyperaldosteronism. Laboratory data showed an undetectable morning ACTH, a morning cortisol ≥ 5.0 $\mu\text{g/dL}$ following 1-mg overnight dexamethasone suppression test on two separate occasions, a normal 24-hour urinary free cortisol (UFC), and a low dehydroepiandrosterone sulfate (DHEAS) (Table 4).

1. Does this adrenal mass represent a malignancy?

No, the imaging characteristics show a lipid-rich mass that is suggestive of benign entity.

The unenhanced CT images show a round right adrenal mass with low attenuation (<10 HU). In addition, the adrenal mass displayed no evidence of local invasion or calcifications. Despite being slightly more than 4 cm in size, the other features of this tumor strongly suggest that it is benign (29,50). Some experts recommend surgical adrenalectomy when lipid-rich adrenal masses are greater than 4 cm in size, owing to concerns that small malignant foci may not be recognized in larger-sized masses; however, this recommendation is not supported by high-grade evidence and should be considered on a case-by-case basis.

2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

Despite the lack of classical physical features on examination suggestive of CS, the biochemical evaluation suggests autonomous hypercortisolism.

Patients with mild or autonomous hypercortisolism typically do not have overt manifestations of CS on physi-

cal examination (51,52). The prevalence of autonomous hypercortisolism with incidentally discovered adrenal masses has been reported as approximately 5% among more than 2,000 patients from 13 different studies (12); however, milder degrees of hypercortisolism may exist in a much higher percentage of individuals with adrenal tumors (30,53,54). An overnight 1-mg dexamethasone suppression test is recommended as the initial test to evaluate for hypercortisolism (55). The measurement of a morning dexamethasone, along with the cortisol value, may not be necessary on every occasion but should be considered when there is concern of patient nonadherence with dexamethasone intake, increased clearance/metabolism of dexamethasone, or poor absorption of dexamethasone. A postdexamethasone cortisol value <1.8 $\mu\text{g/dL}$ is considered to represent a “nonfunctional” tumor. False-positive results are not uncommon and may occur in patients receiving medications that accelerate dexamethasone metabolism (such as phenytoin, carbamazepine, rifampin, and phenobarbital), increase corticosteroid-binding globulin (such as estrogen and mitotane), or in patients with depression and alcohol use (56). A postdexamethasone cortisol value ≥ 5.0 $\mu\text{g/dL}$ is regarded to represent “autonomous” cortisol secretion. Postdexamethasone cortisol levels between 1.8 and 5.0 $\mu\text{g/dL}$ indicate “possible” hypercortisolism, or milder degrees of hypercortisolism, where additional investigations may be helpful (Table 3).

In such circumstances, detection of any of the following results may provide additional support for autonomous, and ACTH-independent, hypercortisolism: low or suppressed ACTH (<10 pg/mL), in addition to low DHEAS, and elevated late-night salivary cortisol or 24-hour UFC (9,57). It is important to underscore the detection of ACTH-independent hypercortisolism when implicating the adrenal mass; nonsuppressed ACTH levels indicate an ACTH-dependent hypercortisolism and attention should be paid to a hypothalamic-pituitary or ectopic

Table 4
Biochemical Testing Results for Case 3

	Pre-adrenalectomy	Postadrenalectomy
ACTH (pg/mL) (8 AM) (reference range: 15-60 pg/mL)	<10	51
Cortisol ($\mu\text{g/dL}$) (8 AM)	9.9	12.79
DHEAS ($\mu\text{g/dL}$) (reference range: 40-100 $\mu\text{g/dL}$)	34	64
Cortisol ($\mu\text{g/dL}$) after overnight 1-mg dexamethasone suppression test	5.20 5.57 (repeated after 3 months)	NA
24-h urinary free cortisol ($\mu\text{g}/24$ h) (reference range, 4.0-50.0 $\mu\text{g}/24$ h)	40.1 μg	NA
HbA1c (%)	6.0%	5.5%
Cholesterol (mg/dL)	236	182
Abbreviations: ACTH = adrenocorticotrophic hormone; DHEAS = dehydroepiandrosterone sulfate; HbA1c = glycated hemoglobin; NA = not available.		

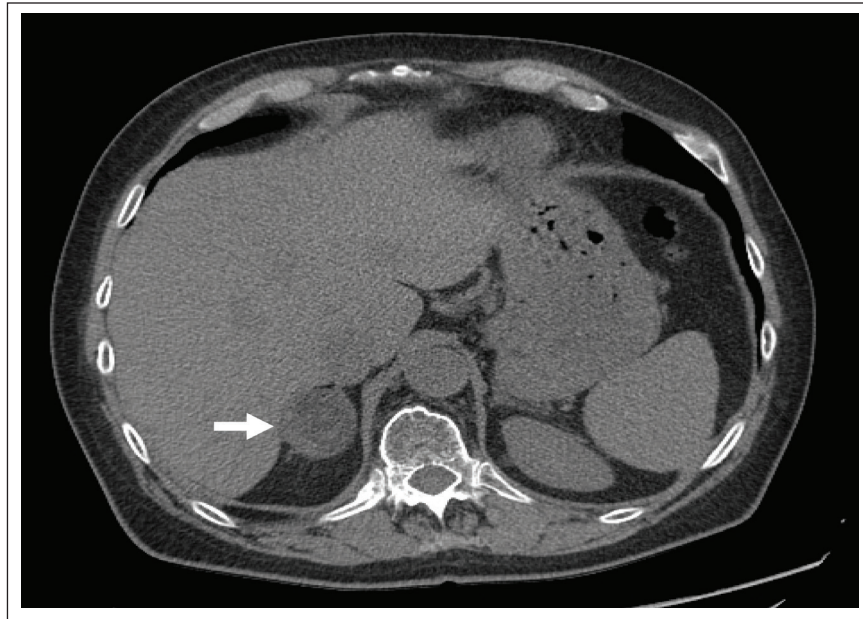


Fig. 4. Noncontrast axial computed tomography image of Case 3 showing a well-circumscribed 4.6 × 3.5 × 3.5 cm lipid-rich (<10 Hounsfield units) right adrenal mass with no evidence of calcifications or local invasion.

source of ACTH. When ACTH values are inconsistent with expectations or the clinical context, it may be reasonable to consider falsely elevated values due to assay interference (58). The dexamethasone suppression test and late-night salivary cortisol test are most sensitive at detecting subtle forms of nonphysiologic and autonomous cortisol secretion, whereas the 24-hour UFC is frequently normal in mild autonomous cortisol excess. The dexamethasone suppression test may also be repeated a few months later to evaluate for consistently elevated results.

In this case, the patient has multiple lines of evidence suggestive of autonomous and ACTH-independent hypercortisolism and clinical evidence associated with cortisol excess (fragility fractures, dyslipidemia, and prediabetes). These collective results may suggest a causal relationship between the hypercortisolism and the adverse clinical outcomes that is amenable to treatment.

3. Would a biopsy of the adrenal mass help with diagnosis or management?

There is no role for adrenal biopsy, as imaging characteristics reveal a benign adrenal adenoma. There is no indication of a malignancy or infiltrative adrenal disease.

4. Is there an indication for surgical or medical treatment?

Surgery should be considered for patients with overt hypercortisolism and CS. For patients with mild or autonomous hypercortisolism without clinical

evidence of CS, an individualized approach for surgery must be taken depending on the presence of cortisol-related comorbidities, patient preference for surgery versus medical monitoring, and other individual patient- and tumor-related factors.

There are no robust randomized controlled studies evaluating the efficacy and value of adrenalectomy for mild or autonomous hypercortisolism. The decision to pursue surgery is based on interpretations of observational data. Observational data from cross-sectional and cohort studies have suggested that autonomous hypercortisolism without overt clinical manifestations of CS is associated with a higher risk for cardiovascular disease, diabetes, hypertension, dyslipidemia, obesity, and fragility fracture (30-33). In parallel, several case series and small uncontrolled intervention studies have suggested that adrenalectomy may improve blood pressure, glycemic control, skeletal health, and weight loss (55,59-61).

After discussing the option of surgical adrenalectomy versus close medical monitoring for new or progressive cortisol-related adverse clinical outcomes, this patient chose to proceed with adrenalectomy. Three months following his surgery, he had a normal (nonsuppressed) ACTH, and his hemoglobin A1c was lower.

When surgical adrenalectomy is pursued for autonomous hypercortisolism, it is often recommended that perioperative glucocorticoid treatment be administered, since it is difficult to assess which patients will develop clinically significant adrenal insufficiency postoperatively (13,62). In a systematic review of the literature evaluating 28 different studies, the prevalence of some degree of adrenal

insufficiency following surgery for autonomous hypercortisolism was approximately 65% (63). Glucocorticoids can be gradually tapered following the surgery depending on the patient's recovery. As an alternative strategy, patients can safely undergo surgical resection of their tumor and have their cortisol levels measured on the morning of the first postoperative day (64). An early morning or random cortisol >15 µg/dL provides reassurance that the function of the contralateral adrenal gland is relatively normal and obviates the need for glucocorticoid therapy (65). Patients with a morning cortisol level <10 µg/dL should be treated with supplemental glucocorticoid therapy with gradual tapering of doses until their hypothalamic-pituitary-adrenal function normalizes. Morning cortisol levels in the 10 to 15 µg/dL range may need glucocorticoid therapy and should be monitored closely to make this determination.

5. Is there an indication for longitudinal surveillance with imaging and/or biochemical testing? If so, how frequently and for what duration?

Yes, if surgery is not pursued, then annual clinical assessment for new or worsening cortisol-related comorbidities such as low bone density, diabetes, weight gain, hyperlipidemia, and hypertension should be performed. The added value of repeating dexamethasone suppression testing annually in a patient with known autonomous hypercortisolism is not clear (62).

CASE 4: A 71-year-old man with a history of hyperlipidemia and prediabetes presented to the endocrine clinic with incidentally discovered bilateral lipid-rich adrenal masses when he underwent a CT scan for back pain.

The right adrenal mass measured $1.7 \times 1.4 \times 1.5$ cm, and the left adrenal mass measured $1.2 \times 0.9 \times 1.0$ cm (Fig. 5). There were no radiologic features suspicious for malignancy, such as heterogeneity, irregular borders, or local invasion. The noncontrast CT attenuation values for the right and left adrenal masses were -10 and -5 HU, respectively. Physical examination was unremarkable. The patient was not overweight and had no supraclavicular fat pads, striae, or proximal muscle weakness. Bone mineral density was normal. Biochemical evaluation was negative for a pheochromocytoma and primary aldosteronism. A morning cortisol was 14.8 µg/dL, with an ACTH of 26 pg/mL and a 24-hour UFC of 10.1 µg. However, a postdexamethasone cortisol was 2.5 µg/dL (see Table 5).

1. Does this adrenal mass represent a malignancy?

No. The imaging characteristics of both adrenal masses are consistent with benign lipid-rich adenomas.

Incidentally discovered adrenal masses may be unilateral or bilateral. Observational series suggest that bilateral adrenal masses can be detected in ~14% of cases and that the left adrenal mass is usually larger than the right, whereas unilateral left-sided masses are detected in ~65% of cases and unilateral right-sided masses in ~21% of cases (66). It is speculated that incidental adrenal masses are more frequently detected on the left rather than the right adrenal gland because small abnormalities on the right adrenal gland are harder to appreciate given its proximity to the liver and kidney. Bilateral adrenal masses may represent nonfunctional adenomas, pheochromocytoma, congenital adrenal hyperplasia, macronodular cortical adenomas that

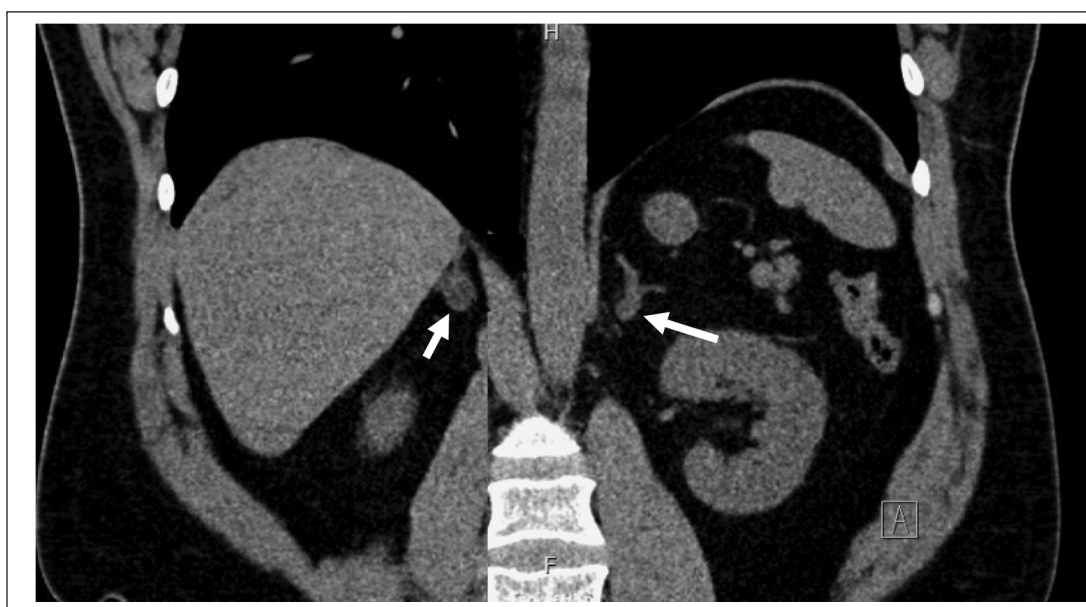


Fig. 5. Noncontrast coronal computed tomography image of Case 4 showing bilateral lipid-rich small adrenal masses (arrows).

Table 5
Biochemical Testing Results for Case 4

	Results
ACTH (pg/mL) (8 AM) (reference range 10-60 pg/mL)	26
Cortisol (μg/dL) (8 AM)	14.8
Cortisol (μg/dL) after overnight 1-mg dexamethasone suppression test	2.5
24-h urinary free cortisol (μg/24 h) (reference range, 4.0-50.0 μg/24 h)	10.1
HbA1c (%)	5.7
Cholesterol (mg/dL)	220
Abbreviations: ACTH = adrenocorticotrophic hormone; HbA1c = glycated hemoglobin.	

cause hypercortisolism (primary macronodular adrenal hyperplasia) or primary aldosteronism, adrenal metastases, lymphomas, and/or infections.

2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

The cortisol following overnight dexamethasone administration was slightly elevated at 2.5 μg/dL, consistent with mild or possible autonomous hypercortisolism. The clinical significance of this biochemical abnormality in this patient is unclear; the relation of this potential hypercortisolism with the patient's prediabetes and dyslipidemia is uncertain.

Although the cortisol level postdexamethasone did not suppress as low as is typically desired, the 24-hour UFC was normal (suggesting the lack of overt hypercortisolism), the morning ACTH concentration was not suppressed (arguing against autonomous and ACTH-independent pathophysiology), and unlike Case 3, there were no other overt clinical manifestations to suggest hypercortisolism. Prior observational studies have indicated that even this mild degree of hypercortisolism may contribute to a higher incident risk of cardiometabolic outcomes (31,32); however, whether a medical or surgical intervention here could mitigate this risk is unclear. Thus, the finding of mild hypercortisolism may serve as a prognostic risk factor until additional studies (i.e., randomized controlled trials) provide more concrete evidence on how best to manage them.

3. Would a biopsy of the adrenal mass help with diagnosis or management?

There is no indication for a biopsy.

4. Is there an indication for surgical or medical treatment?

There is no strong evidence to support any treatment at this time.

An individualized treatment plan is recommended for patients with evidence for autonomous or subclinical hypercortisolism (Fig. 2). Unlike in Case 3, this patient has no visible clinical manifestations of hypercortisolism and no suppression of ACTH. Observational data suggest that surgical adrenalectomy may have beneficial cardiometabolic effects in some instances of autonomous cortisol excess (60); however, until high-grade evidence from randomized intervention studies are available, strong recommendations for this patient cannot be made, and an individualized approach is required.

When treatment is warranted in patients with bilateral adrenal tumors and autonomous hypercortisolism, several options could be considered, including observation and repeated testing at a later time (in the absence of overt comorbidities associated with hypercortisolism or a suppressed ACTH), medical therapy to decrease cortisol secretion or action, unilateral adrenalectomy of the larger adrenal mass, or bilateral adrenalectomy. Adrenal vein sampling for cortisol and metanephrines (at specialized and experienced centers) may be considered in select cases with bilateral adenomas of similar size prior to considering surgery (67-69).

5. Is there an indication for longitudinal surveillance with imaging and/or biochemical testing? If so, how frequently and for what duration?

Annual clinical assessment for signs and symptoms of hypercortisolism, with repeated dexamethasone suppression testing to assess progression in the severity of hypercortisolism, should be considered.

CASE 5: *A 65-year-old man with a history of depression and osteoarthritis was found to have a lipid-poor adrenal mass that was 5.0 × 4.5 × 5.0 cm on a CT scan performed to evaluate back pain. The mass was lobular and heterogeneous, with a noncontrast CT attenuation of 55 HU. (Fig. 6). Physical exam was unremarkable. He was a tall, thin man without supraclavicular fat pads, striae, or proximal muscle weakness. Biochemical evaluation*

revealed normal plasma metanephrines and aldosterone-to-renin ratio. However, the patient had an abnormal dexamethasone suppression test, with a posttest cortisol of 8 µg/dL. DHEAS was elevated at 550 µg/dL (normal, 40 to 100 µg/dL).

1. Does this adrenal mass represent a malignancy?

The imaging characteristics of this adrenal mass are concerning for adrenal malignancy.

The large size, heterogeneous content, lobular contours, and lipid-poor density on unenhanced CT imaging are all concerning for a malignant entity. The elevated DHEAS and abnormal dexamethasone suppression results further indicate a high likelihood of primary adrenocortical carcinoma that is secreting cortisol and adrenal androgens.

2. Is there evidence for clinically relevant and autonomous adrenal hormone excess?

There is no overt physical evidence for adrenal hormone excess. However, the patient had an abnormal dexamethasone suppression test and elevated DHEAS. The combination of glucocorticoid and androgen cosecretion is highly suggestive of adrenocortical carcinoma (53).

Adrenal cortical carcinoma can present with clinical and/or biochemical hyperandrogenism or hypercortisolism (70); however, they may also present as hormonally silent. Less frequently, adrenocortical carcinoma may present

with hyperaldosteronism or estradiol excess. Patients who develop adrenal cortical carcinoma tend to have rapid tumor growth. As a result, in the case of a secreting tumor, clinical manifestations can also develop and progress rapidly (71).

3. Would a biopsy of the adrenal mass help with diagnosis or management?

A biopsy is not indicated.

If there is suspicion that a patient has adrenocortical carcinoma based on radiographic and/or biochemical findings, then a biopsy is not indicated. The main reason to avoid biopsy is that adrenocortical carcinomas are sufficiently heterogeneous that focal sampling may not be diagnostic, and in some cases, may falsely suggest a benign diagnosis (42). A biopsy may be useful for diagnosing extra-adrenal metastases or invasive adrenal infections after a pheochromocytoma has been excluded.

4. Is there an indication for surgical or medical treatment?

Yes, adrenalectomy should be performed by a surgeon with a high level of experience in performing adrenal cancer surgeries.

When the concern for adrenocortical carcinoma arises, the most appropriate diagnostic, prognostic, and therapeutic intervention is an open and radical adrenalectomy. This intervention offers the ability to obtain gross pathology that

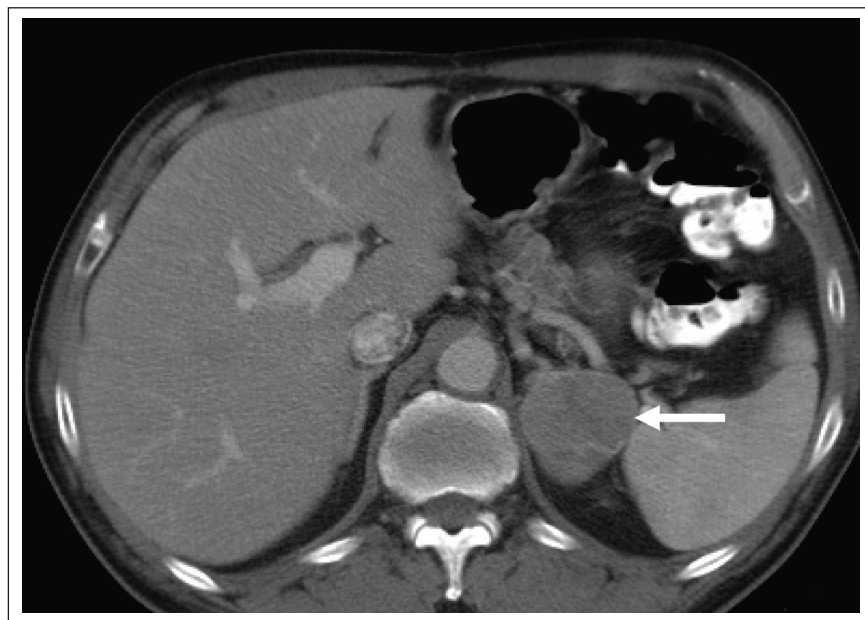


Fig. 6. Postcontrast transverse computed tomography image of Case 5 showing a 5.0 × 4.5 × 5.0 cm lipid-poor (>10 Hounsfield units on noncontrast computed tomography) heterogeneous left adrenal mass and lobular contour.

can confirm the diagnosis as well as the grade and stage of the tumor and also permits removal of the tumor along with exploration of lymph nodes. Currently, the only cure for adrenocortical carcinoma is a complete surgical resection, which also remains the best therapy to prolong survival time (72,73). This patient should be treated by a multidisciplinary team including dedicated experts in endocrinology, endocrine surgery, medical oncology, radiation oncology, and pathology. If such expertise is not available locally, referral to the nearest adrenal center is advised to ensure the best outcome.

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DISCLOSURE

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